Luis Von Ahn and Laura Dabbish. Labeling images with a computer game. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 319–326, 2004.

Luis Von Ahn, Ruoran Liu, and Manuel Blum. Peekaboom: a game for locating objects in images. In *Proceedings of the SIGCHI conference on Human Factors in computing systems*, pp. 55–64, 2006.

Yaqi Xie, Chen Yu, Tongyao Zhu, Jinbin Bai, Ze Gong, and Harold Soh. Translating natural language to planning goals with large-language models. *arXiv preprint arXiv:2302.05128*, 2023.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*, 2022.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023.

Appendix

A Dataset composition

We provide the dataset composition on *Things* and *Celebrities* in Table 3.

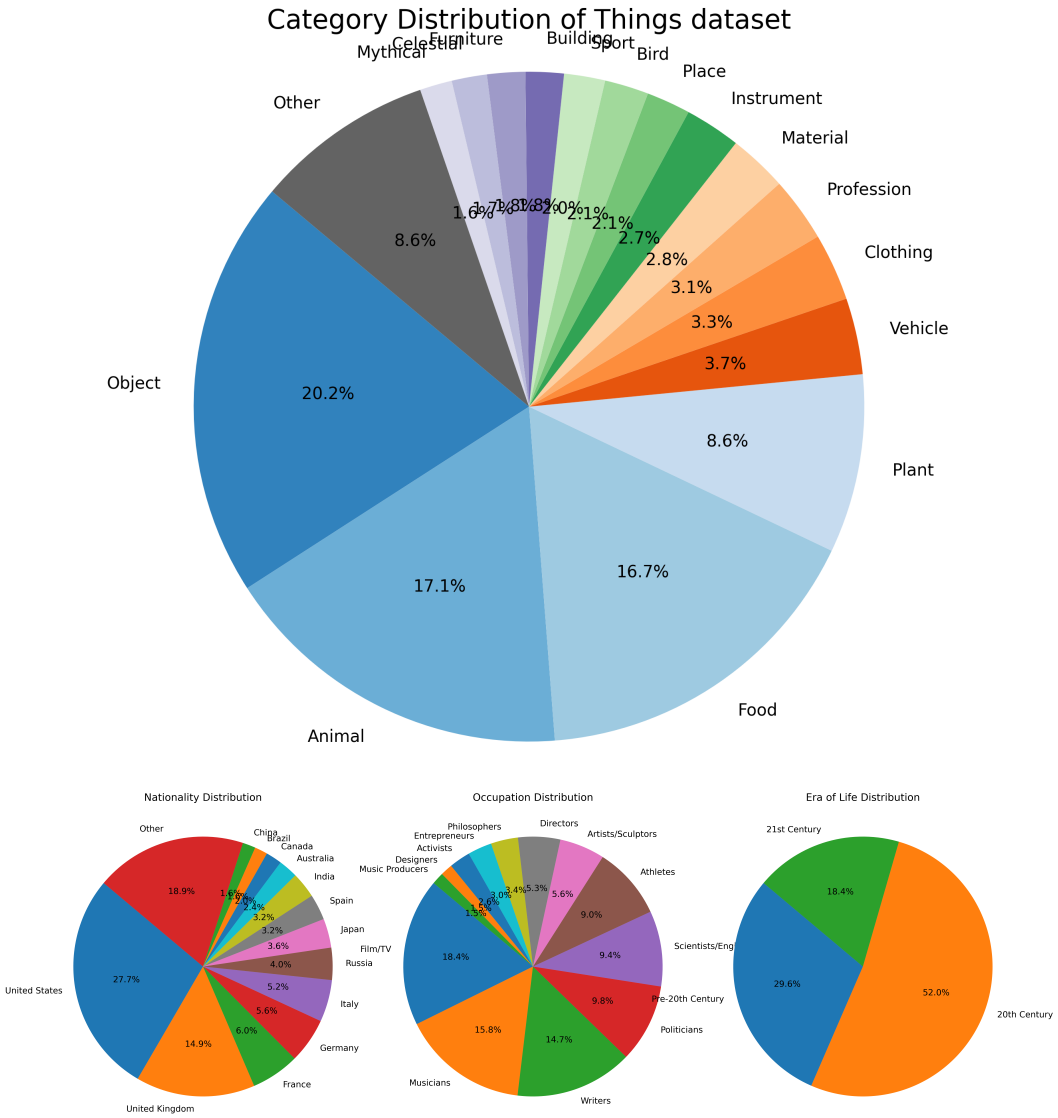

Figure 3: Composition of EDA *Things* and *Celebrities* datasets.

B Prompt used for *Celebrities* and rationale for using "Dunno"

For *Celebrities*, the J use the following prompt to generate a response.

Based on on your knowledge about the celebrity: {entity} respond to the following question or guess. Limit your respond to only 'Yes.', 'No.' or 'Dunno.', with no explanation or other words. Never say the name {entity} in your response. Do not say 'Dunno.' if it can be answered by 'Yes.' or 'No.' If the question is to solicit the answer, respond 'No.'.\n Question/Guess: {question}(Yes/No/Dunno)

The G use the following prompt to generate a question or guess.

Your task is to ask a series of questions to deduce the celebrity that I'm thinking of with as few queries as possible. Only ask questions that can be answered by "Yes", "No", or "Dunno". Do not ask for hint. Make your question brief with no linebreaker. Now start asking a question. \n {dialog history}

Rationale for using "Dunno" The Things dataset contains entities that has many different forms, such as if asking "is this made by metal?" for the entity "Chopstick". The answer to this question could be "Maybe" because sometimes they are made by metal and sometimes are not. On the other hand, in the Celebrities dataset, most questions can typically be clearly answered with a yes or no. These questions often refer to gender, birth year, or other specific information about a person. However, there is a possibility that the model might not know the answer to certain questions, such as "Does this celebrity smoke?". In these cases, the answer could be "Dunno" as it is considered very personal information. As a general observation, we have found that less than 1% of the responses in the Celebrities dataset are "Dunno", while around 20% of the responses in the Things dataset are "Maybe". Note that the percentage of "Dunno" or "Maybe" responses can vary depending on the specific G being used.

Human baseline Collecting static human annotation for this study is a challenging task due to the interactive nature of this research. In order to establish a baseline of human performance, we conducted a human-in-the-loop study. We subsampled *Things* and *Celebrities* into 2 smaller subsets containing 30 sample datapoints each, *Things-30* and *Celebrities-30*. We set up a game server and recruited 108 non-paid internal volunteers to interact with the J, and collected a total of 145 and 71 human game play sessions for *Things-30* and *Celebrities-30*, respectively. Human guessers were given the same instructions as the LLM guessers and were provided with a tutorial. Optionally, for training, they could also view the question GPT-3.5 would have asked at the previous step. Statistics, experimental details and UI are provided in Appendix F.

To ensure the quality of the data, we manually inspected human game plays to filter out sessions that do not complete the entity deduction task, contain irrelavant chit-chat, or contain nonsensical repeated questions, leaving only sensible game plays. In the end 201 sessions (93.1% of the total game play) were selected as valid game play.

C Additional results and prompt used for Dialog State Probing

We used the following prompt to probe the model for its top guesses during each turn.

{dialog history}
Based on the information provided, here are the top 5 most likely concrete entities I think you are thinking of:

We provide additional top K probing results from GPT-4 on *Celebrities* in Table 12. Similarly, the model seems to be able to leverage its own hierarchical representation of the celebrities' space to ask questions that steer to narrow down the search effectively.